

	Work Instruction	Doc. Revision	00
	AXI V810 STD Outer Barrier Valve & Speed Controller Upgrade (S/N:3073 & Below)	Effective Date	2 Oct 2016

This procedure is applicable to below models:

V810 STD	☒	S1	☒	S2
V810 XXL	☐	S1	☐	S2
V810 S2EX	☐			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	2 Oct 2016	First Release	ONG KEAN SHENG

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	AXI V810 STD Outer Barrier Valve & Speed Controller Upgrade (S/N:3073 & Below)	Effective Date	2 Oct 2016

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Parts requirement :

Part Number	Part Description	Qty
2860-0003	TUBING, TU POLYURETHANE, 6MM	1 Roll
2820-0117	FITTING,FLOW CONTROL,D6.0	2
4XASP-A107	Outer barrier Solenoid Valve kit	1

Parts removal

1. Close SSCA
2. Close X ray Source at Operator Console Panel
3. Unplug Parker Clip Fitting as below Figure 1.

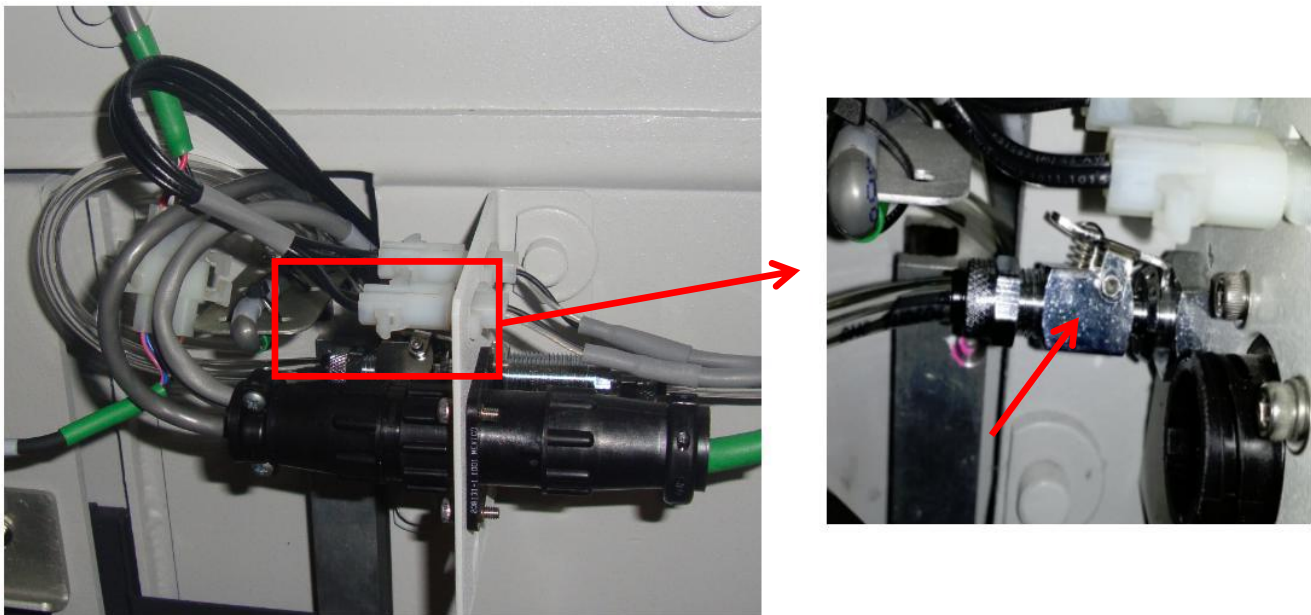


Figure 1

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4. Remove 2 x tubing from outer barrier speed controller. May refer to below Figure 2.



Figure 2

5. Remove 2 x Outer Barrier Speed Controller as below Figure 3



Figure 3

6. Remove 2 x Tubing from Outer Barrier Solenoid Valve (Outlet Port) as below Figure 4.

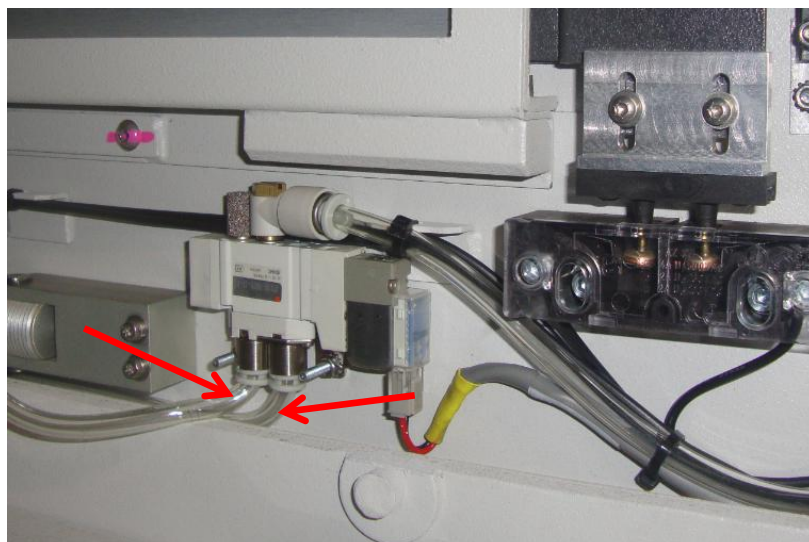


Figure 4

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7. Remove 1 x Tubing from Outer Barrier Solenoid (Inlet Port) as below Figure 5. Disconnect Input cable as well.

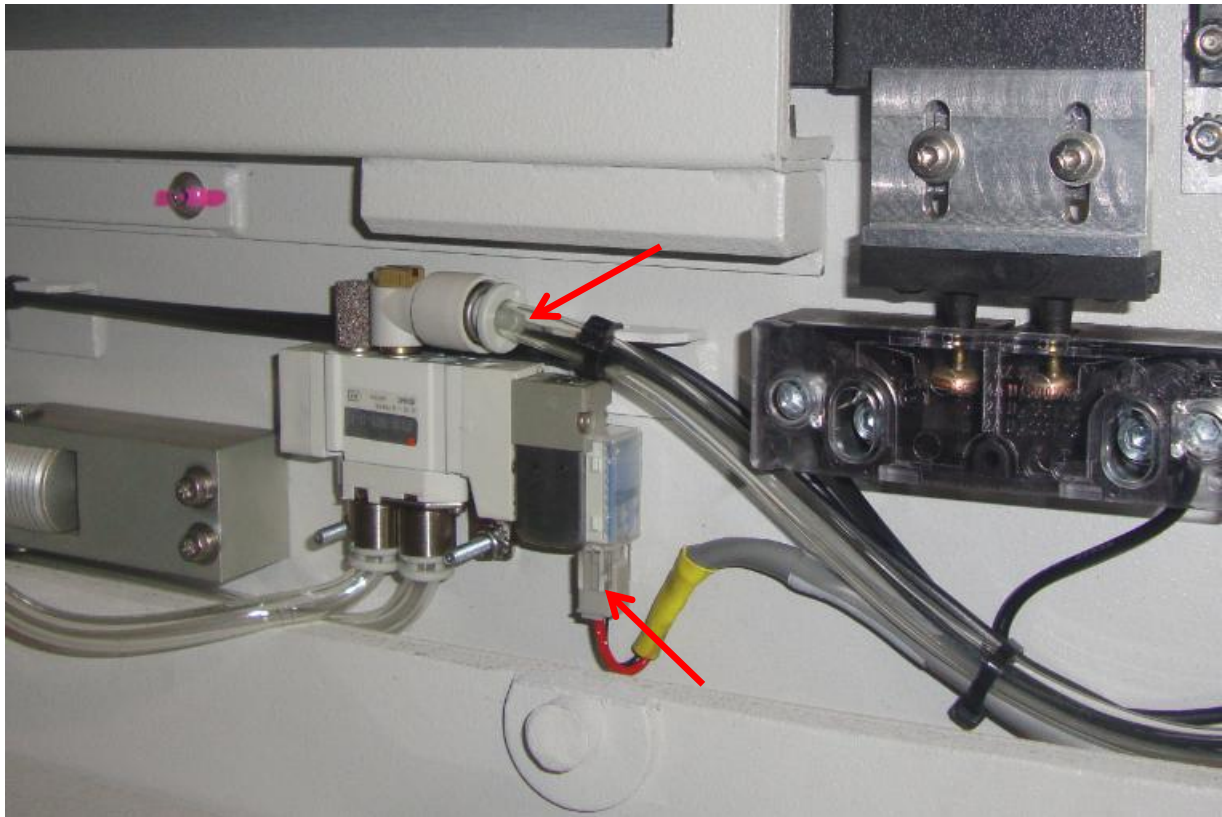


Figure 5

8. Loosen 2 x Nut and remove entire Outer Barrier Solenoid Valve as below Figure 6.

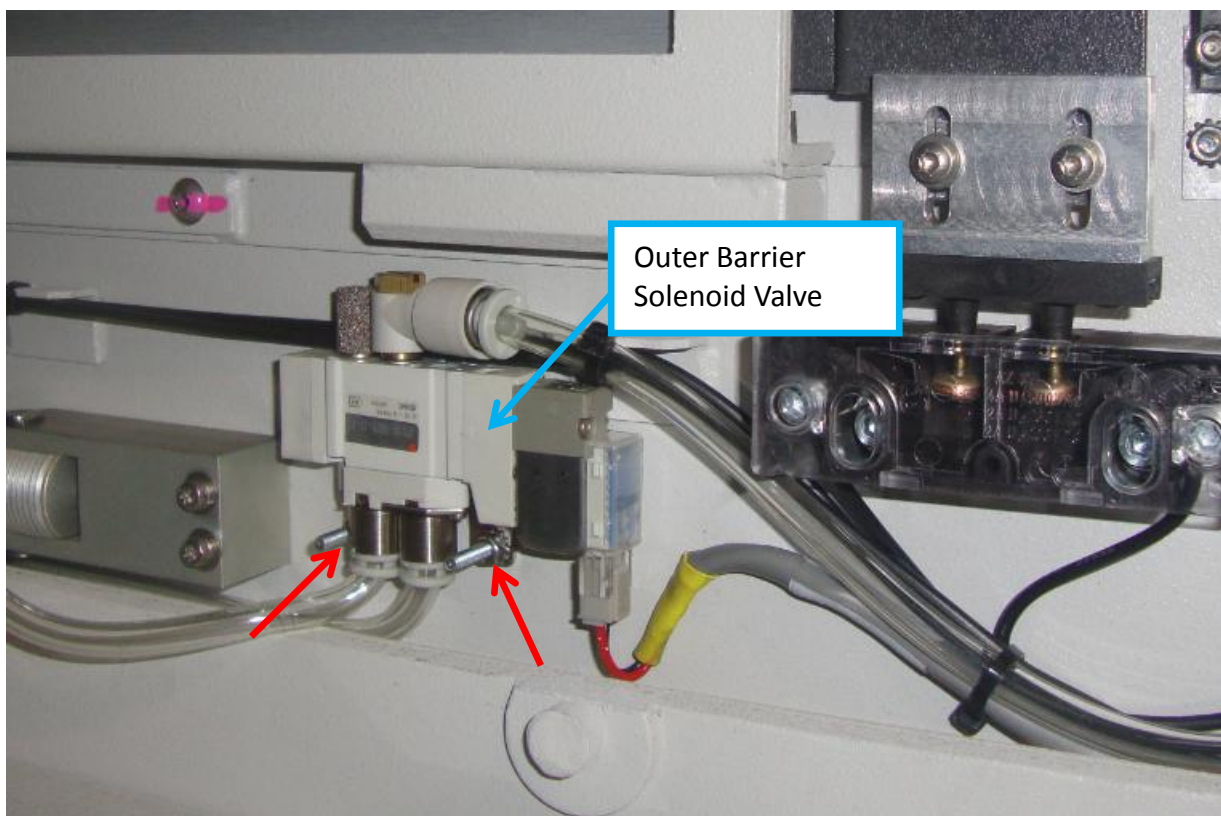


Figure 6

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9. Remove entire tubing together with Parker clip fitting. May refer to below Figure 7.

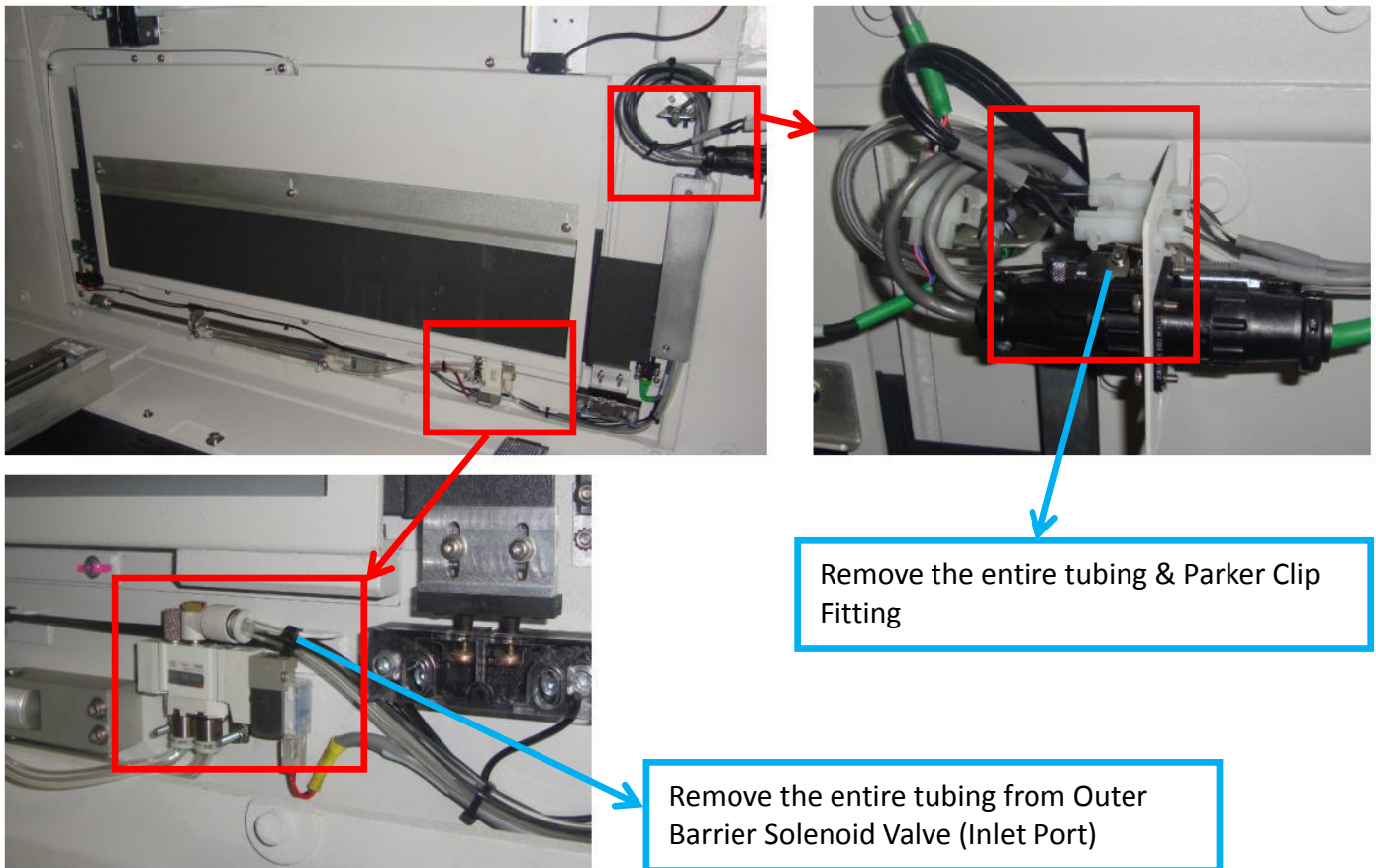


Figure 7

10. Gently loosen the nut and slide through existing tubing. May refer to below Figure 8.

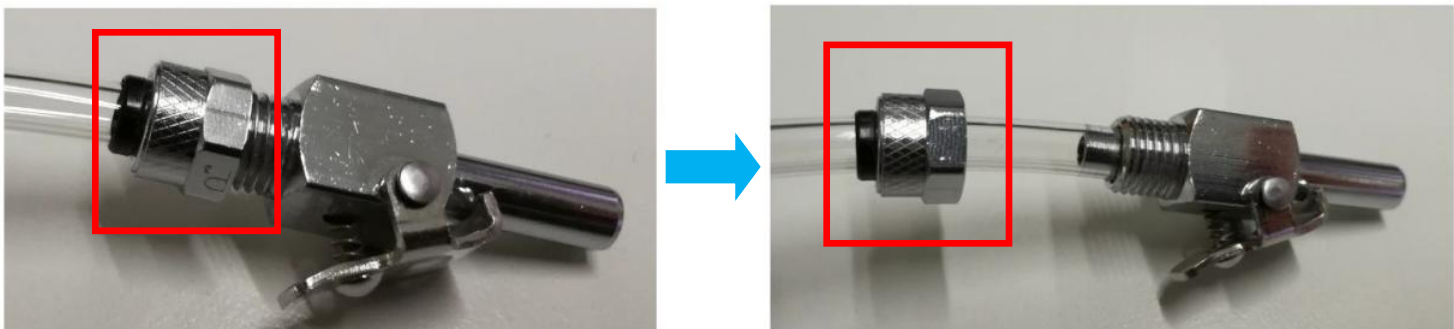


Figure 8

11. Gently remove the existing tubing from Parker Clip Fitting as Figure 9.

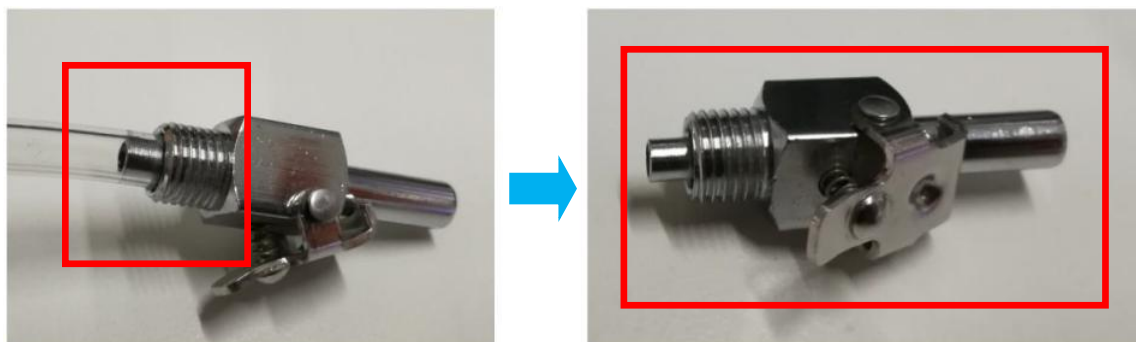


Figure 9

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Part replacement

12. Assemble New Outer Barrier Solenoid Valve Kit (PN : 4XASP-A107) at outer barrier as below Figure 10. Tighten 2 x Nut.



Figure 10

13. Assemble 2 x New Speed Controller (2820-0117) at outer barrier cylinder as Figure 11.

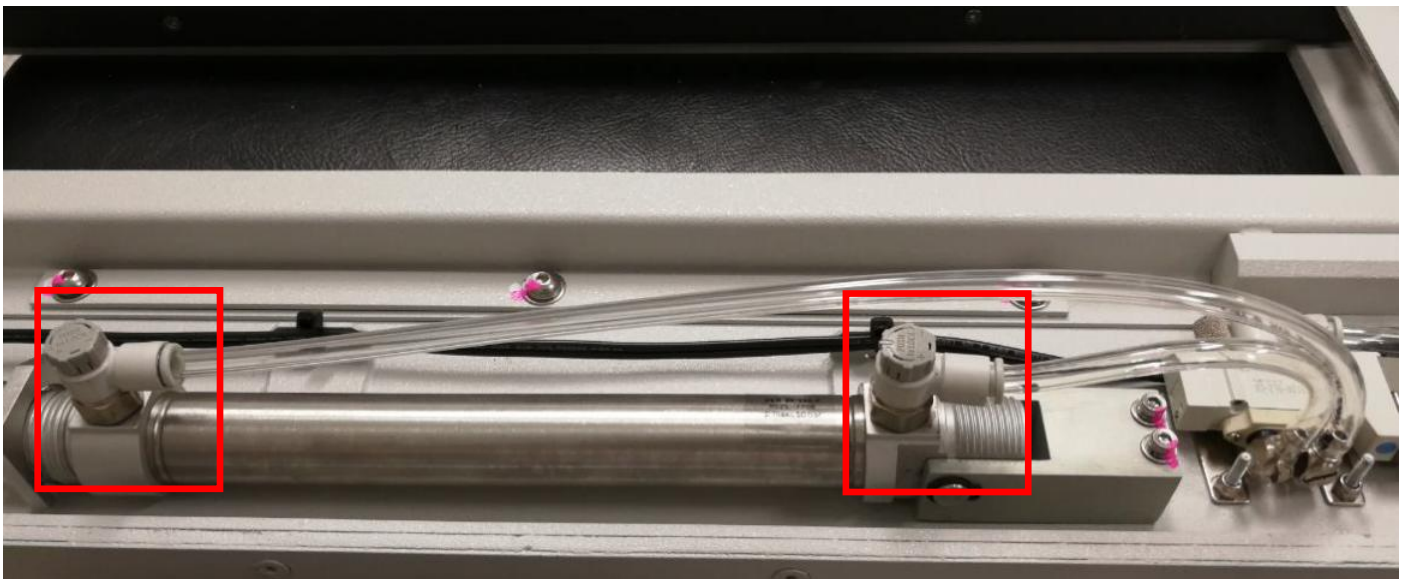


Figure 11

14. Cut 2 x tubing (PN : 2860-0003) into desire length by refer to below Figure 12. Plug 2 x Tubing by refer to below orientation.

Orientation	Routing Path	Cutting Length (Approximately)
A1-A2	Speed Controller to Outer Barrier Valve	28.5cm
B1-B2	Speed Controller to Outer Barrier Valve	10cm

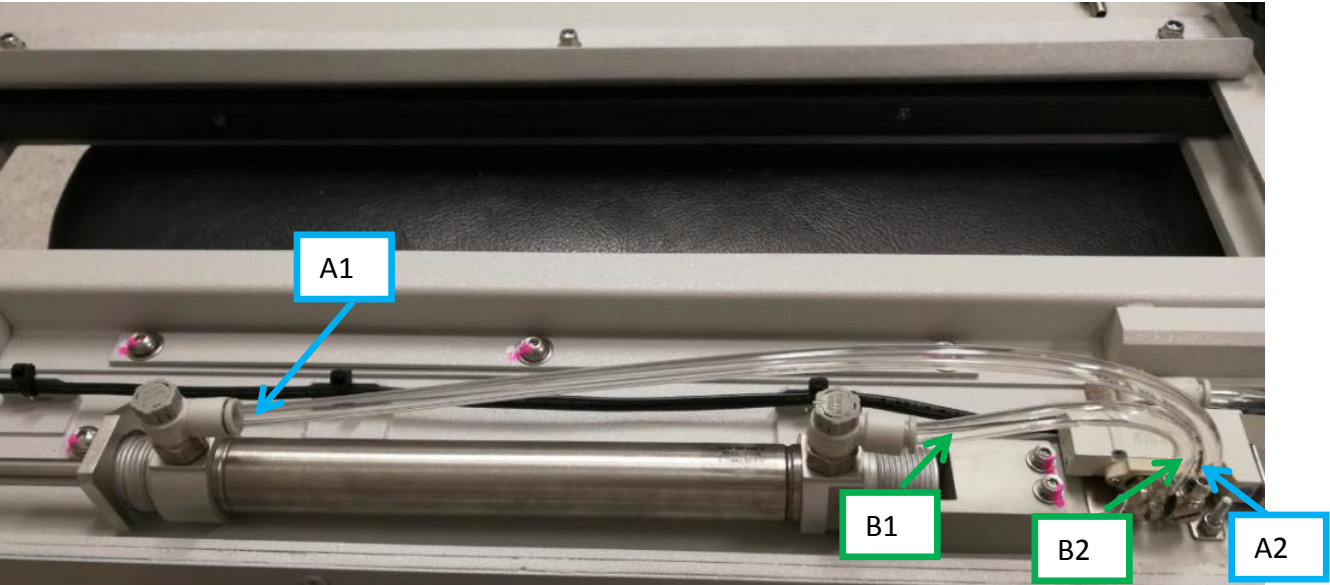


Figure 12

15. Cut 1 x Tubing (PN : 2860-0003) into desire length (Approximately 76cm) by refer to routing path (Blue Highlight)

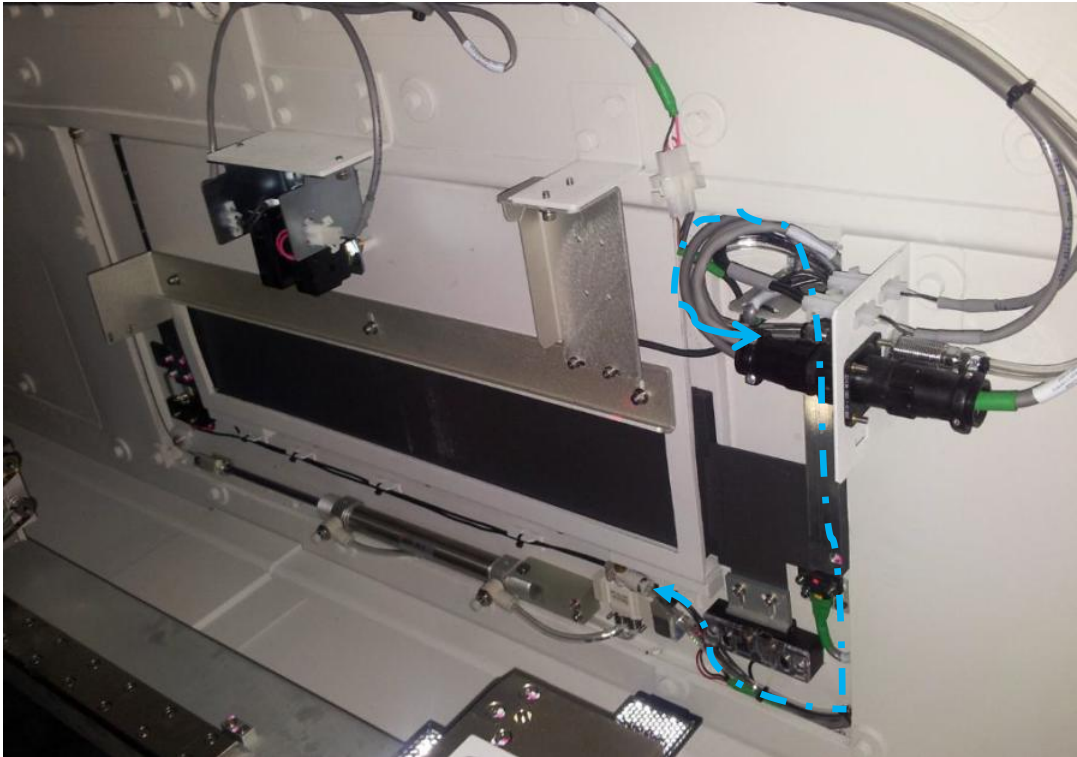


Figure 13

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16. Plug the other end of tubing (Approximately 76mm) at Parker Clip Fitting. May refer to below Figure 14.

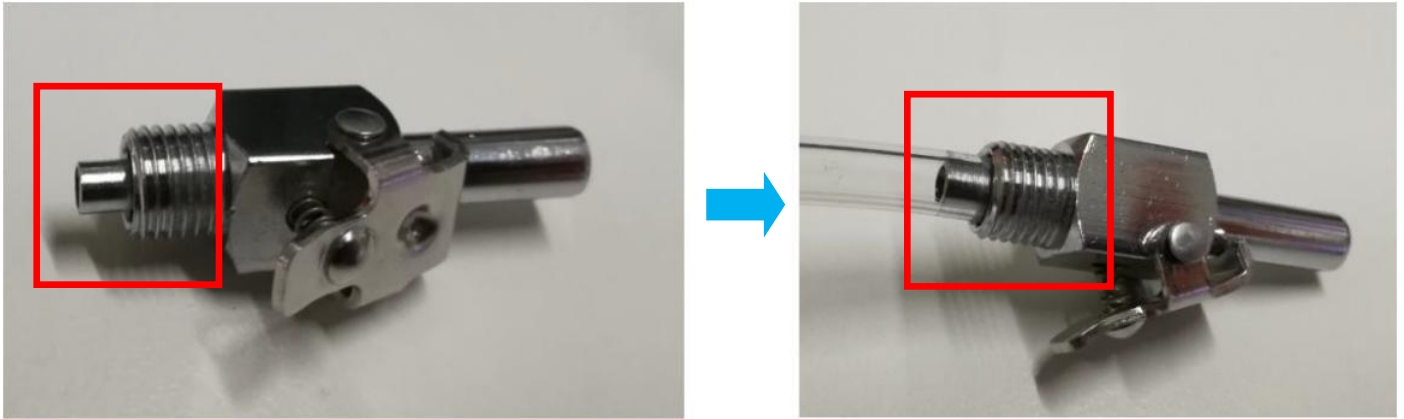


Figure 14

17. Gently slide the nut through tubing. Tighten the nut. May refer to Figure 15.

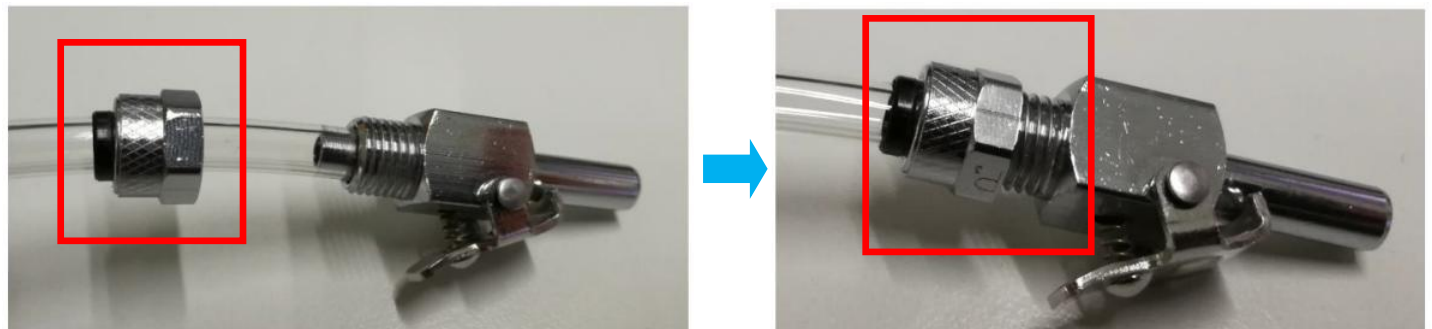


Figure 15

18. Gently plug the other end of tubing into New Outer Barrier Solenoid Valve (Inlet port) & apply cable tie to secure as Figure 16. Connect input cable.

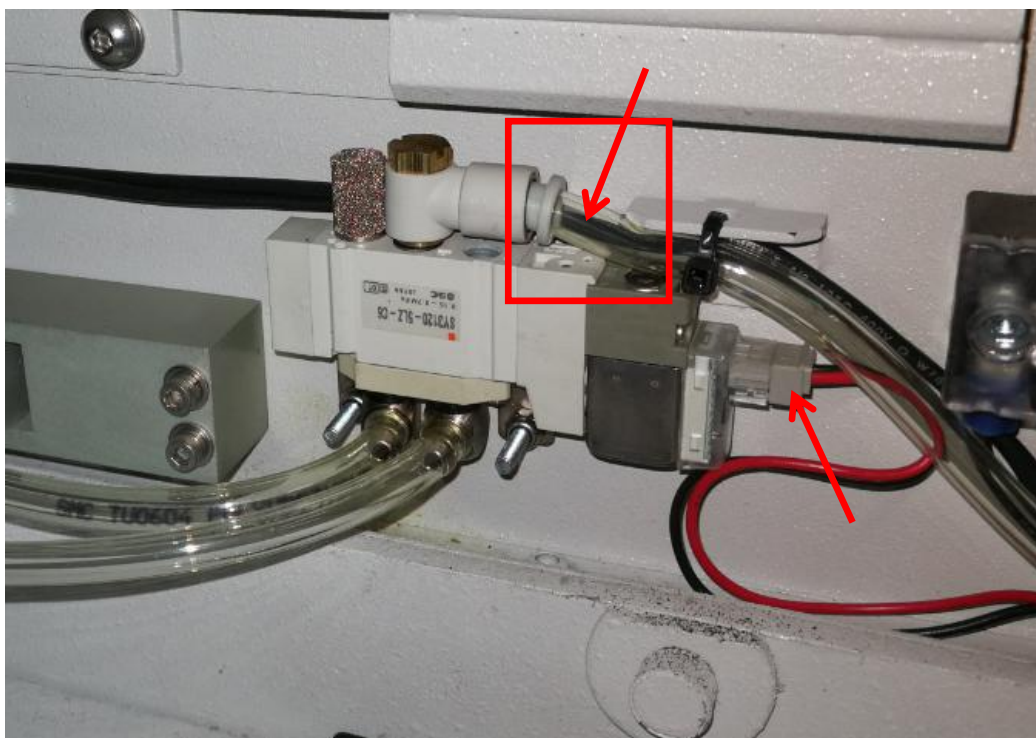


Figure 16

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19. Gently route the entire tubing by refer to below route (Blue Highlight). Press & lock the Parker Clip Fitting. May refer to below Figure 17 & 18. Apply necessary cable ties to secure the tubing.

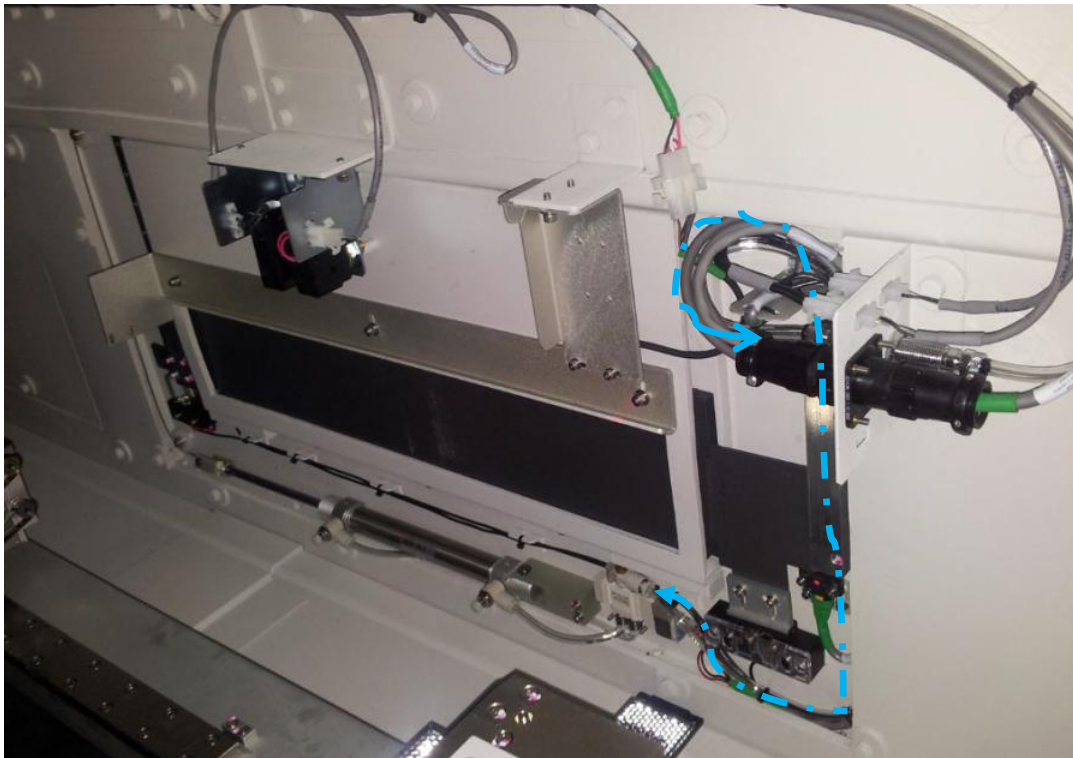


Figure 17

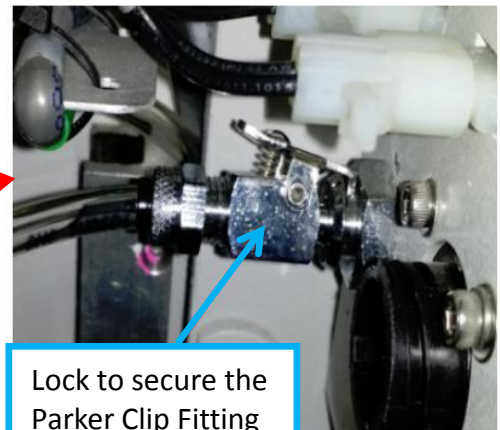
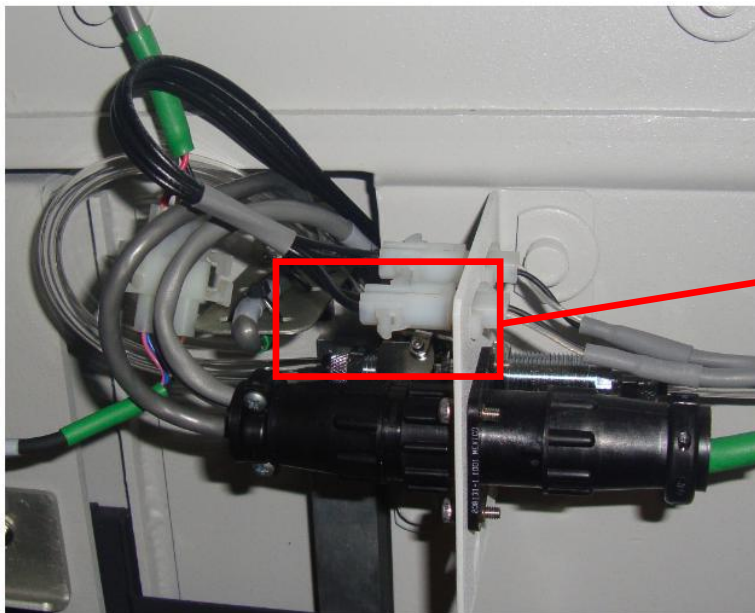


Figure 18

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20. Manually move the X Axis Stage towards particular barrier. May refer to below Figure 19.

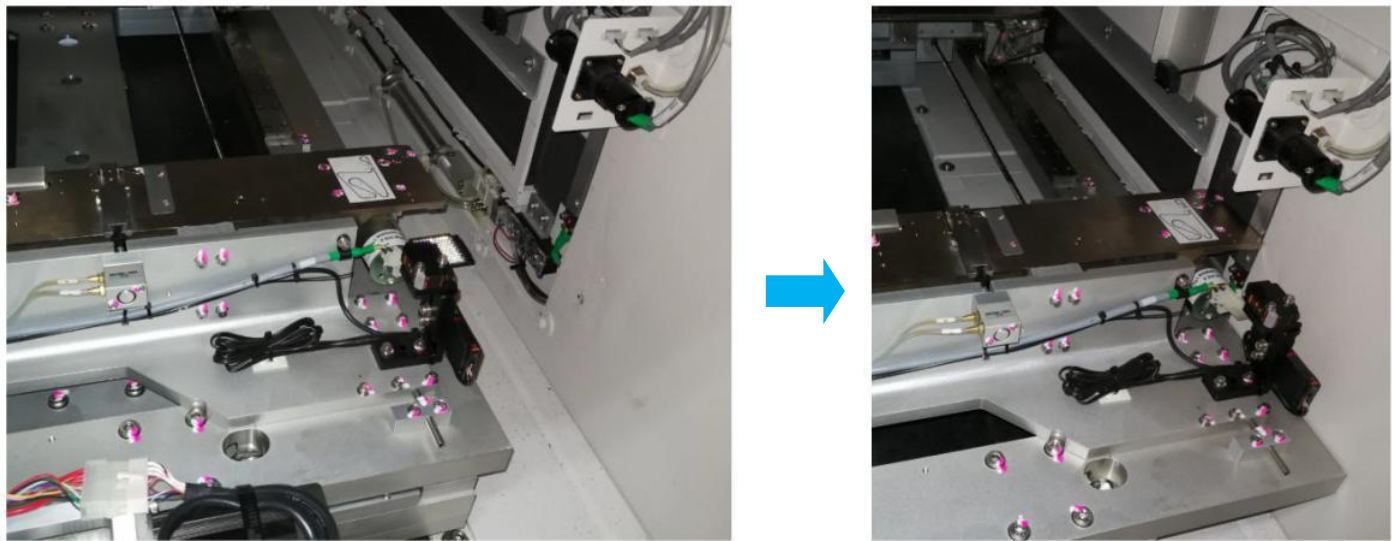


Figure 19

21. Visual check outer barrier tubing whether contain any friction with X Axis Stage.

a) If friction persist, re-arrange tubing position or re-cut new tubing into desire length.

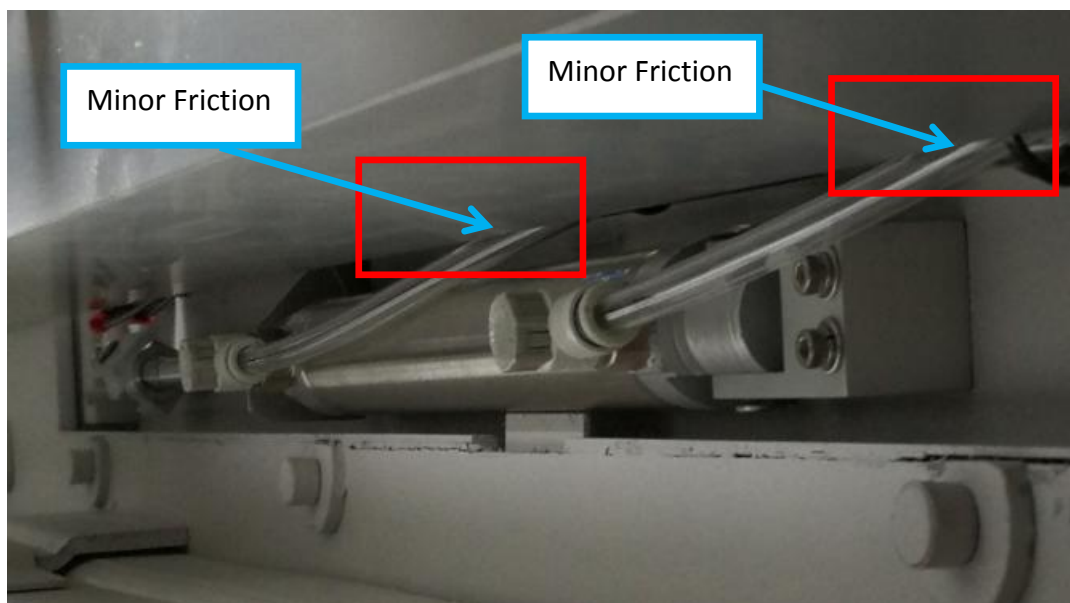


Figure 20

22. Remove all tools from XY Stage

23. Close all access door

24. Open SSCA

25. Open V810 GUI

26. Initialize at Panel handling

27. Open & Close particular Outer barrier which perform Valve & Speed controller upgrading. Record down Opening & Closing duration as below Figure 21.

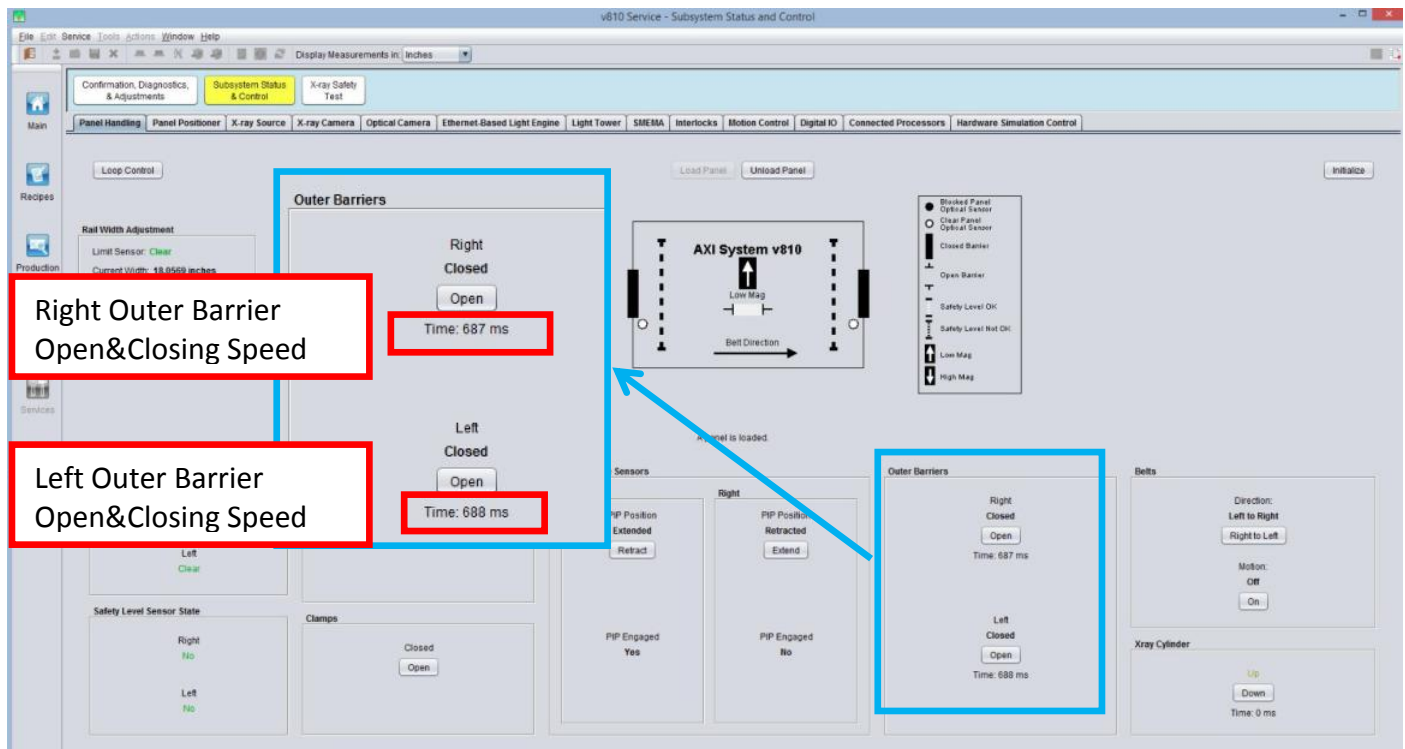


Figure 21 Outer Barrier Opening & Closing Duration

28. Adjust 2 X Speed Controller at Outer Barrier cylinder by refer to below table.

Outer Barrier Status	Outer Barrier Speed
Open	500ms~750ms
Close	500ms~750ms

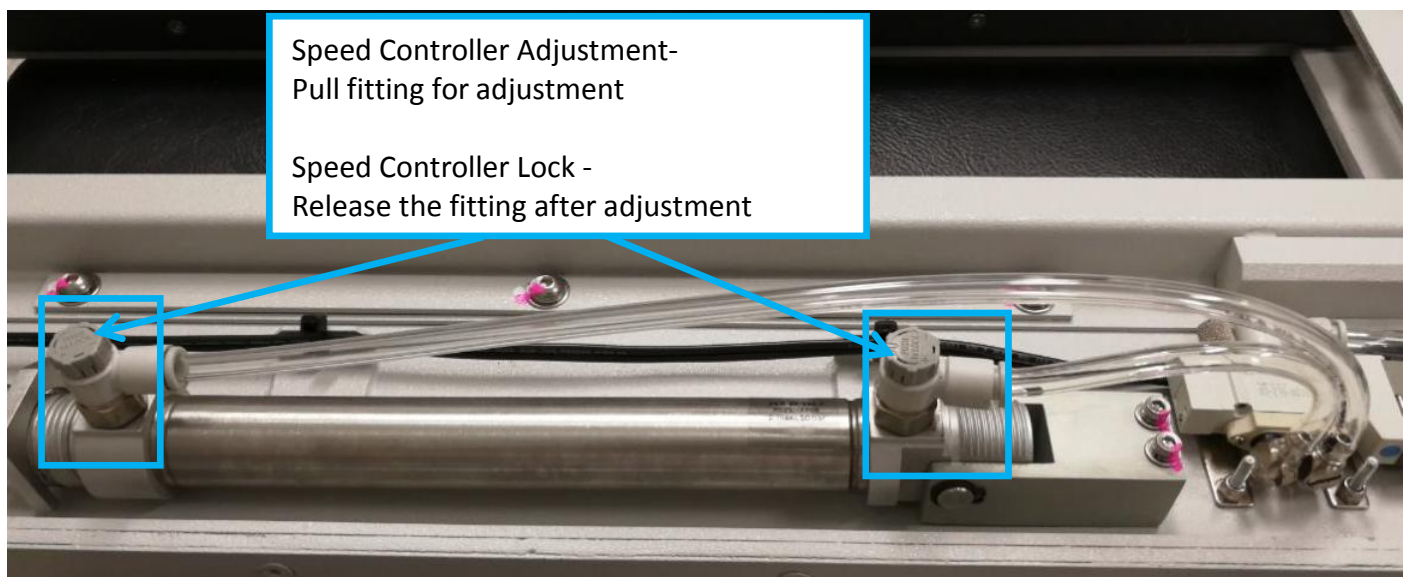


Figure 22